

Forecasting Condiment Farmgate Price Volatility in the MIMAROPA Region Using Supervised Regression and Spatial Clustering

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Abstract— Farmgate price volatility affects the income and decision-making of condiment farmers in the MIMAROPA region. This study developed a machine learning-based framework to forecast farmgate prices and analyze provincial market volatility using Philippine Statistics Authority (PSA) data from 2021 to 2025. The dataset was cleaned and preprocessed before applying Linear Regression to forecast the price trend of Siling Labuyo and K-Means Clustering to classify provincial markets based on average prices and price volatility. Welch's independent two-sample t-test was performed to validate the observed price changes between 2022 and 2025. The results indicated an overall upward trend in farmgate prices, identified three provincial market profiles with varying levels of stability, and confirmed that the price differences between 2022 and 2025 were statistically significant. The findings demonstrate that machine learning techniques can support agricultural price forecasting and provide useful insights for farmers, policymakers, and other agricultural stakeholders in improving market planning and decision-making.

Keywords—Farmgate Price Forecasting, Machine Learning, Linear Regression, K-Means Clustering, Agricultural Price Volatility, MIMAROPA.

I. INTRODUCTION

Agriculture serves as a primary economic driver in the MIMAROPA region, encompassing the provinces of Occidental Mindoro, Oriental Mindoro, Marinduque, Romblon, and Palawan. While rice remains the staple crop, high-value condiments such as garlic, ginger, and chili peppers have emerged as critical sources of livelihood for many farming households. In regions like Occidental Mindoro, farmers strategically plant garlic after the rice harvest to utilize residual soil moisture and rice straw, thereby maximizing the use of agricultural land [3].

Despite their intrinsic economic value, the local condiment sector faces severe systemic challenges. According to the Philippine Statistics Authority (2024), national garlic production stands at merely 5,830 metric tons annually, meeting only 2.6% of national demand. Consequently, the Philippines imports approximately 95% of its garlic supply [3], [9]. Furthermore, domestic yields are comparatively low, averaging 1–2 tons per hectare, compared with the 12–15 tons achieved by nations with advanced agricultural technologies [10].

Local producers also face highly inequitable pricing structures. While urban consumers pay ₱140–₱150 per kilogram for garlic, farmers receive drastically lower farmgate prices. A parallel disparity exists in the ginger market, where farmgate prices can plummet to ₱30 per kilogram, even as Metro Manila retail prices reach ₱280–₱300 per kilogram [9]. This stark disparity indicates that intermediaries capture the majority of the profit margins. Furthermore, the archipelagic geography of MIMAROPA inflates logistical costs. The lack of optimized cold storage and food distribution infrastructure frequently results in localized oversupply and crop spoilage, preventing smallholder farmers from accessing profitable markets despite the sector's contribution to regional economic growth (Philippine Statistics Authority, 2026).

Smallholder condiment farmers in the MIMAROPA region face significant financial instability due to erratic farmgate prices, geographic isolation, and limited access to market information. Environmental factors and unsynchronized harvest schedules often cause farmgate prices to drop dramatically, with some commodities falling to as low as ₱30 per kilogram while retail prices in urban markets can reach ₱300 per kilogram, creating substantial market distortions and reducing farmer income [9]. In addition, fluctuations in consumer demand contribute significantly to price volatility, making it difficult for farmers and other stakeholders to anticipate market conditions and plan their production activities effectively [5]. The region's geographic characteristics also force many farmers to depend heavily on intermediaries and regional wholesalers to sell their produce, weakening their bargaining power and limiting access to real-time market valuations [3]. As a result, the absence of localized, data-driven forecasting tools prevents farmers from making informed decisions and increases their vulnerability to unpredictable market changes.

This study develops a machine learning framework for forecasting farmgate price trends and regional market volatility of high-value condiments in the MIMAROPA region using historical data from the Philippine Statistics Authority. The findings may help farmers make more informed decisions regarding harvesting and marketing activities by providing predictive insights into future price movements, thereby reducing dependence on intermediaries and improving potential income. Agricultural cooperatives and local government units may utilize the results to identify

areas with high market volatility and implement targeted interventions, such as agricultural support programs and storage infrastructure. Policymakers and government agencies may also benefit from the predictive trends generated by the model in addressing supply chain inefficiencies, monitoring price fluctuations, and supporting agricultural stability. Furthermore, the study contributes to the growing body of research on the application of machine learning in agricultural economics and provides a foundation for future studies on predictive analytics in island-based economies such as MIMAROPA.

II. LITERATURE REVIEW

A. Farmgate Prices and Price Volatility

As agricultural commodities remain highly vulnerable to market fluctuations, Catacutan and Ulep examined the determinants of coffee farmgate prices in the Philippines [2]. Utilizing time-series data from 1997 to 2016, the study analyzed the effects of production costs and net returns on farmgate prices through a Vector Autoregression (VAR) model. The findings established significant relationships among these variables, indicating that fluctuations in production costs directly determine the final price farmers receive. Furthermore, the study suggested that improving operational efficiency contributes to price stability and higher producer returns. These findings underscore how baseline economic factors dictate agricultural pricing, thereby justifying the need to analyze historical farmgate behavior in regions like MIMAROPA.

In a closely related context, Hinlo and Lee investigated the primary drivers of price volatility in Philippine condiment markets, focusing on onions and garlic [5]. By analyzing quarterly data from 2010 to 2024, the researchers decomposed price fluctuations into demand, supply, and import components. The results revealed that demand variability and domestic market conditions were the primary drivers of price volatility, playing a much more significant role than international trade factors. Additionally, the study noted that seasonal production patterns heavily intensified price instability. Because garlic and similar condiment crops are central to the present study, these findings highlight the critical importance of utilizing time-series analysis to anticipate farmgate price fluctuations and regional market instability.

B. K-Means Clustering and Machine Learning

A study by Bublik et al. characterized the German grain legume market as fragmented with limited competition due to structural restrictions on farmer market access [1]. Using K-Means clustering for network structuring and optimization, the authors revealed a tight geographic concentration of traders in southern and western Germany, leaving the majority of outlying farmers without viable access points even within a 100-kilometer radius. The study demonstrated that a more competitive market for agricultural producers cannot be achieved by increasing the sheer number of local dealers, but rather by optimizing and expanding the trading distance between farmers and buyers. This spatial reality is a direct parallel to the archipelagic inter-island distribution bottlenecks that smallholders in MIMAROPA face.

In evaluating predictive architectures, Theofilou et al. tested the efficacy of Machine Learning (ML), Ensemble

Learning (EL), Deep Learning (DL), and Time-Series (TS) methods for forecasting baseline prices of staple crops such as wheat, corn, and rice [11]. Their systematic review confirmed that these advanced algorithmic approaches significantly outperform traditional econometric models, particularly regarding predictive consistency and long-term reliability.

Conversely, the study emphasized that while these advanced forecasting methods are highly effective at safeguarding food security, they remain hampered by persistent operational hurdles. Specifically, these models are frequently constrained by high computational demands, low interpretability, and restricted generalizability due to their rigid over-reliance on localized, region-specific training data. Furthermore, the absence of standardized benchmark datasets across the agricultural sector makes it exceptionally difficult to cross-reference model performance or replicate historical findings [11].

Ultimately, both studies illustrate that while advanced data mining is a vital mechanism for modernizing agriculture, algorithms alone cannot resolve systemic supply chain deficits. Machine learning can successfully identify physical market bottlenecks and map future price trajectories, but its practical efficacy remains bound to the physical architecture of the trade network. To construct a genuinely secure and competitive agricultural landscape, the deployment of predictive analytics must be coupled with fundamental structural reforms, ranging from expanding physical trading infrastructure to standardizing the fragmented data generated by regional farmgate economies.

III. METHODOLOGY

This study employed a machine learning-based methodology to analyze farmgate price trends and market volatility in the MIMAROPA region. The methodology consisted of three major phases: data collection and preprocessing, machine learning integration, and statistical inference. Python was used as the primary programming language, and all analyses were performed using Google Colab.

A. Data Collection and Preprocessing

The dataset used in this study was obtained from the Philippine Statistics Authority (PSA) OpenSTAT database. The extraction focused on farmgate prices of selected condiments across the five provinces of the MIMAROPA region from January 2021 to December 2025. The resulting dataset contained 5,040 records organized in a wide-format structure.

To prepare the dataset for analysis, exploratory data analysis and preprocessing were performed using the Pandas library. The original wide-format dataset was transformed into a tidy format through data un pivoting. Records containing null values and invalid price entries were removed to improve data quality and prevent model bias. Date values were converted into a time-series format, and a numerical month index was generated to support forecasting operations.

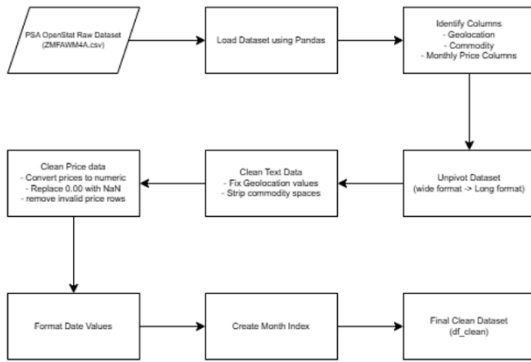


Fig. 1. Data Collection and Preprocessing Workflow

B. Linear Regression Forecasting

A supervised learning approach was utilized to forecast future pricing trajectories. Using the scikit-learn library, a Linear Regression model was trained on historical Siling Labuyo pricing data. Time was treated as the independent variable, and the farmgate price was the dependent variable. This model quantified the exact monthly inflation rate and projected the trendline into the following year.

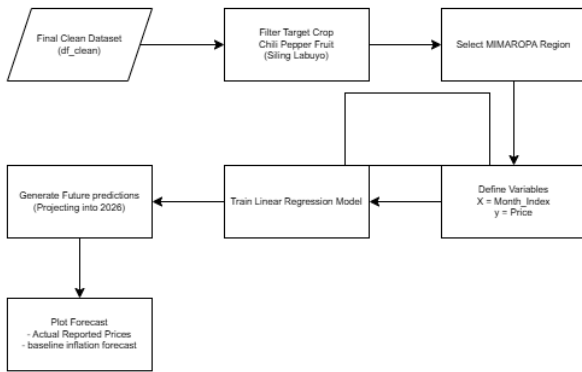


Fig. 2. Linear regression forecasting workflow

C. Market Categorization via K-Means Clustering

An unsupervised learning algorithm was deployed to evaluate regional market stability. The K-Means algorithm analyzed the average price and price volatility of the five specific provinces. The data was first standardized using a scaler function to ensure mathematical accuracy. The provinces were then grouped into distinct risk profiles, such as High Volatility or Stable Markets.

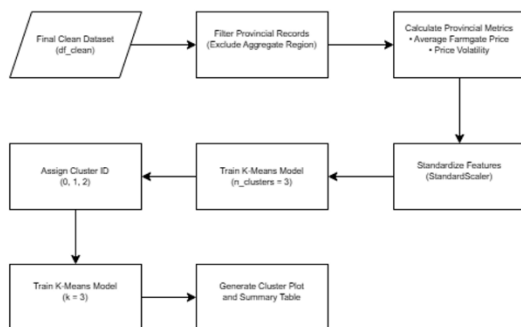


Fig. 3. K-Means Clustering Workflow

D. Statistical Inference

To validate the machine-learning findings, the study assessed the statistical significance of the observed price inflation. An independent two-sample t-test was executed using the SciPy library.

During the initial testing phase, the data revealed that the Philippine Statistics Authority did not record valid farmgate prices for Siling Labuyo in 2021. To maintain strict data integrity, the baseline year was dynamically adjusted. The T-test compared the recorded prices from 2022 with those in 2025.

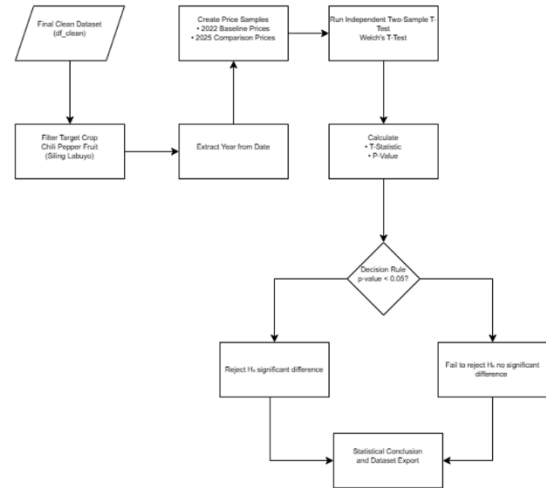


Fig. 4. Statistical Inference Workflow

IV. RESULTS AND DISCUSSION

This chapter presents the outputs generated by the machine learning algorithms and statistical tests applied to the Philippine Statistics Authority farmgate price dataset. The analysis evaluates regional market volatility and establishes predictive trajectories for agricultural commodities in the MIMAROPA region.

A. Data Preprocessing Results

The preprocessing process greatly improved the quality of the datasets before implementing machine learning techniques. Initially, there were 5,040 farmgate price data from January 2021 to December 2025. Data exploration showed that 3,189 pieces of data have zero values and 720 pieces of data have missing values. This is because these data occurred when the product was not produced or there were no farmgate prices. In order to make the data more complete, these records were removed resulting in 1,131 pieces of data to be used in the study. Moreover, the baseline year for Siling Labuyo was changed from 2021 to 2022 since there were no valid farmgate prices in 2021.

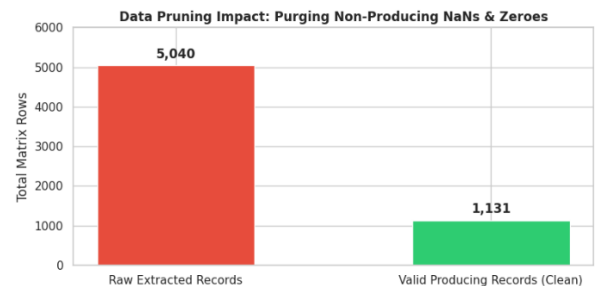


Fig. 5. Data Pruning Impact: Purging Non-Producing NaNs and Zeroes

Figure 5 illustrates the effect of data preprocessing on the dataset. The reduction in records reflects the removal of invalid observations while preserving valid production data. This preprocessing step minimized bias caused by missing or zero-value records and produced a reliable dataset for machine learning and statistical analysis.

B. Exploratory Data Analysis

1. Provincial Farmgate Price Distribution

To estimate future farmgate price behavior, a Linear Regression model was trained using the cleaned Siling Labuyo dataset. The model identified a positive relationship between time and farmgate price, indicating an overall upward trend throughout the observation period. The computed regression coefficient of 1.53 suggests that the average farmgate price increased by approximately ₱1.53 per kilogram per month, reflecting a gradual increase in prices over time.

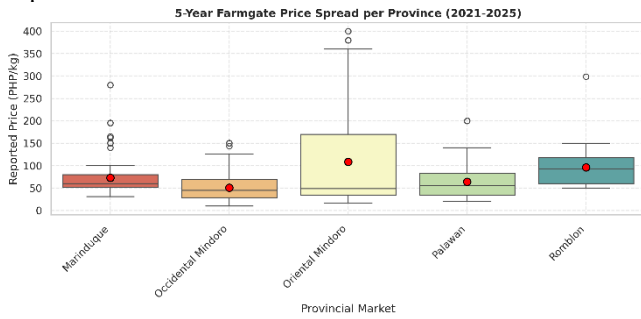


Fig. 6. Five-Year Farmgate Price Spread per Province (2021–2025)

Figure 6 shows that farmgate prices varied considerably across provinces. Some provinces exhibited relatively narrow price ranges, indicating more stable market conditions, whereas others displayed wider distributions and several extreme values, suggesting higher price volatility. These differences indicate that market behavior is not uniform across the region and support the use of clustering techniques to categorize provincial market conditions.

2. Historical Farmgate Price Trends

Historical price trajectories were analyzed to observe long-term trends and seasonal changes in Siling Labuyo farmgate prices across the provinces.

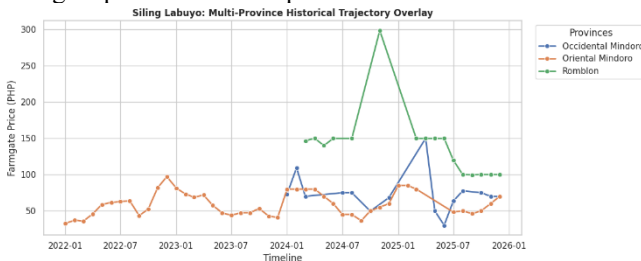


Fig. 7. Siling Labuyo Multi-Province Historical Trajectory Overlay

Figure 7 illustrates the monthly farmgate price movements from 2021 to 2025. Although all provinces experienced fluctuations throughout the study period, the magnitude and timing of these changes differed. Some provinces maintained relatively consistent prices, while

others experienced sharp increases and decreases during specific periods, reflecting the influence of seasonal production, market demand, and regional supply conditions.

C. Linear Regression Forecasting Results

To estimate future farmgate price behavior, a Linear Regression model was trained using the cleaned Siling Labuyo dataset. The model identified a positive relationship between time and farmgate price, indicating an overall upward trend throughout the observation period. The computed regression coefficient of 1.53 suggests that the average farmgate price increased by approximately ₱1.53 per kilogram per month, reflecting a gradual increase in prices over time.

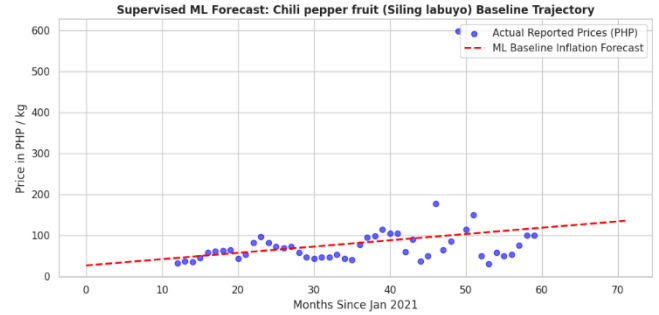


Fig. 8. Supervised ML Forecast: Chili Pepper Fruit (Siling Labuyo) Baseline Trajectory

Figure 8 presents the historical farmgate prices together with the fitted regression line and projected price trend. The forecast indicates that the upward trajectory is expected to continue beyond the observed period, suggesting persistent inflationary pressure in the MIMAROPA condiment market if current conditions remain unchanged.

D. K-Means Clustering Results

K-Means clustering was applied to classify provinces according to average farmgate prices and price volatility. Using three clusters, the model identified distinct market profiles representing varying levels of market stability across the region.

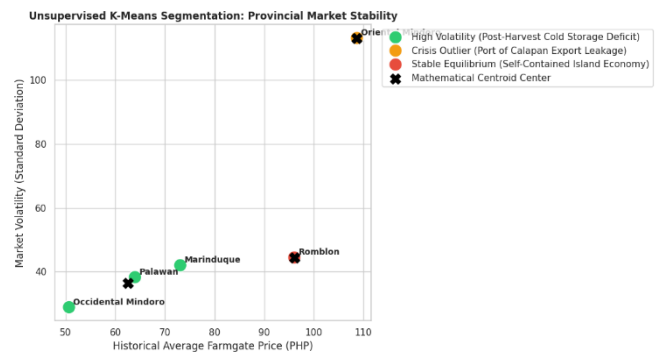


Fig. 9. Unsupervised K-Means Segmentation: Provincial Market Stability

The clustering results indicate that Romblon belongs to the stable market cluster because of its relatively low average prices and limited price variation. Palawan, Marinduque, and Occidental Mindoro were grouped into the high-volatility cluster, reflecting greater fluctuations in farmgate prices over time. Oriental Mindoro formed a separate cluster characterized by both the highest average

prices and the greatest price volatility, indicating a unique market condition compared with the other provinces. These findings demonstrate that provincial geography and local market conditions contribute to differences in farmgate price stability throughout the MIMAROPA region.

E. Statistical Validation

To determine whether the observed increase in farmgate prices represented a statistically significant change, Welch's independent two-sample t-test compared valid Siling Labuyo prices recorded in 2022 and 2025. The analysis used 24 observations from 2022 and 39 observations from 2025. The computed t-statistic was -2.8670, with a corresponding p-value of 0.0063.

Table I. Results of Welch's Independent Two-Sample T-Test

Metric	Value
Sample Size (2022)	24
Sample Size (2025)	39
T-statistic	-2.8670
P-value	0.0063
Decision	Reject H_0

Since the p-value is lower than the significance level of 0.05, the null hypothesis is rejected. This result indicates that the difference in average farmgate prices between 2022 and 2025 is statistically significant rather than the result of random variation. The statistical findings support the machine learning models by confirming that the observed increase in farmgate prices reflects a meaningful structural change in the MIMAROPA agricultural market.

V. CONCLUSION

This study developed a machine learning-based approach for forecasting condiment farmgate price volatility in the MIMAROPA region using historical farmgate price data from the Philippine Statistics Authority. Through data preprocessing, linear regression forecasting, K-Means clustering, and statistical validation, the proposed methodology successfully analyzed historical pricing patterns and identified significant differences in provincial market behavior.

The data preprocessing stage improved dataset quality by removing invalid and non-producing records, resulting in a reliable dataset for analysis. Linear Regression identified an overall upward trend in the farmgate prices of Siling Labuyo, suggesting that prices may continue to increase if current market conditions persist. K-Means clustering classified the provinces into distinct market profiles based on average farmgate prices and price volatility, demonstrating that provincial markets exhibit different levels of stability. Furthermore, Welch's independent two-sample t-test confirmed that the difference between the 2022 and 2025 farmgate prices was statistically significant, supporting the conclusion that the observed increase in prices represents a meaningful market change rather than random variation.

Overall, the findings demonstrate that machine learning techniques can provide valuable support for agricultural price forecasting and market analysis. The proposed approach can assist farmers, agricultural cooperatives, and government agencies in understanding price behavior, anticipating market fluctuations, and making more informed production and marketing decisions.

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